

Cybersecurity Training Diary

Day 3: Basic Networking Concepts (IP, TCP/UDP, DNS, HTTP/HTTPS)

1. Topics Covered

✓ IP (Internet Protocol) Addresses

- **Purpose:** Unique identifier for devices on a network.
- **Types:**
 - **IPv4:** 32-bit (e.g., 192.168.1.1), running out of addresses.
 - **IPv6:** 128-bit (e.g., 2001:0db8:85a3::8a2e:0370:7334), solves IPv4 exhaustion.
- **Public vs. Private IPs:**
 - **Public:** Used on the Internet (assigned by ISP).
 - **Private:** Used internally (e.g., 192.168.x.x, 10.x.x.x).

✓ TCP vs. UDP

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented (handshake)	Connectionless (no handshake)
Reliability	Guaranteed delivery (retransmits lost packets)	No guarantees (faster)
Use Cases	Web browsing (HTTP), emails	Video streaming, gaming (DNS, VoIP)

✓ DNS (Domain Name System)

- **Purpose:** Translates human-readable domain names (e.g., google.com) to IP addresses.
- **How it Works:**
 1. Your computer asks a **DNS resolver** (e.g., Google's 8.8.8.8).
 2. Resolver checks **root servers** → **TLD servers (.com)** → **Authoritative servers**.
- **Security Risks:** DNS spoofing (attacker redirects traffic to malicious sites).

✓ HTTP vs. HTTPS

Feature	HTTP (Hypertext Transfer Protocol)	HTTPS (HTTP Secure)
---------	------------------------------------	---------------------

Feature	HTTP (Hypertext Transfer Protocol)	HTTPS (HTTP Secure)
Security	No encryption (plaintext)	Encrypted (TLS/SSL)
Port	Port 80	Port 443
Use Case	Legacy websites (risky)	Secure transactions (banks, logins)

2. Key Takeaways

- IP Addresses are like phone numbers for devices.
- TCP = Reliable but slow | UDP = Fast but unreliable.
- DNS is the Internet’s phonebook—but can be hacked.
- HTTPS encrypts data (look for in browser).

3. Practical Exercises

Task 1: Find Your IP & DNS Server

- Windows: `ipconfig /all`
- Linux/macOS: `ifconfig` or `ip a`
- Check DNS: `nslookup google.com`

Task 2: Compare HTTP vs HTTPS

- Open Wireshark (packet analyzer).
- Visit `http://example.com` and `https://example.com`.
- Observe unencrypted (HTTP) vs encrypted (HTTPS) traffic.

4. Questions for Reflection

- Why does DNS matter for cybersecurity? (Hint: What if a hacker changes `bank.com` to their IP?)
- When would you use UDP instead of TCP?